

DEL RIO, TX, USA

WMO: 722610

Lat: 29.378N

Lon: 100.927W

Elev: 305

StdP: 97.72

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 22010

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.3	(c) 1.4	(d) -12.7	(e) 1.3	(f) 10.1	(g) -10.3	(h) 1.6	(i) 10.7	(j) 11.4	(k) 15.2	(l) 9.6	(m) 13.8	(n) 2.3	(o) 330	(p) 0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.5	(c) 39.1	(d) 22.4	(e) 38.1	(f) 22.5	(g) 37.1	(h) 22.6	(i) 25.5	(j) 32.3	(k) 25.1	(l) 31.8	(m) 24.7	(n) 31.2	(o) 5.2	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.6	27.4	23.4	18.9	27.3	22.9	18.4	27.2	80.1	32.1	78.0	31.7	76.3	31.2	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.3	7.6	-4.1	41.2	2.0	1.4	-5.5	42.2	-6.7	43.0	-7.8	43.8	-9.2	44.8
			-5.5	26.6	1.9	1.3	-6.9	27.5	-8.0	28.3	-9.0	29.0	-10.4	30.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	11.7	14.5	18.4	22.5	26.3	29.5	30.4	30.6	27.3	22.3	16.3	11.8
	DBStd	7.75	4.36	4.96	4.51	3.79	3.20	2.19	2.12	2.01	2.86	3.98	4.35	4.13
	HDD10.0	85	33	16	3	0	0	0	0	0	0	1	6	27
	HDD18.3	705	209	123	56	10	1	0	0	0	1	13	88	205
	CDD10.0	4403	84	141	265	374	506	584	633	637	518	382	194	84
	CDD18.3	1981	2	16	59	134	248	334	375	379	268	136	26	3
	CDH23.3	23753	41	172	507	1299	2755	4358	5111	5181	2982	1150	166	32
	CDH26.7	12406	6	47	161	539	1355	2408	2966	3021	1468	402	28	4
Wind	WSAvg	3.9	3.2	3.6	4.1	4.5	4.5	4.7	4.4	3.9	3.4	3.6	3.3	3.1
Precipitation	PrecAvg	483	14	24	22	47	57	57	41	37	60	58	25	16
	PrecMax	844	49	96	88	191	175	146	335	155	251	288	85	78
	PrecMin	235	0	0	0	0	2	0	0	0	0	0	0	0
	PrecStd	134	13	23	24	42	37	46	60	38	56	64	23	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	31.3	33.1	36.7	39.4	40.5	40.4	40.1	39.0	34.6	29.4	26.5
		MCWB	14.9	15.4	16.2	18.6	20.4	22.4	22.7	22.9	21.9	21.6	18.6	15.7
	2%	DB	24.2	27.8	30.7	33.9	36.8	38.2	38.9	38.7	36.6	32.3	27.1	23.8
		MCWB	14.0	14.9	17.0	19.5	21.8	22.6	22.6	22.8	22.4	21.5	18.1	15.0
	5%	DB	21.9	25.3	28.3	31.8	34.9	36.8	37.8	37.7	34.9	30.7	25.2	21.5
		MCWB	13.0	15.1	16.9	19.3	21.5	22.7	22.7	22.8	22.4	21.0	17.5	14.4
	10%	DB	19.7	22.7	26.2	29.8	33.1	35.5	36.7	36.7	33.3	29.0	23.4	19.4
		MCWB	12.7	14.2	16.8	18.6	21.7	22.7	22.7	22.8	22.2	20.5	16.9	13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	20.0	21.4	23.6	25.5	26.5	25.8	25.7	25.6	24.9	22.4	19.6
		MCDB	22.3	23.9	26.1	29.7	31.8	33.3	33.4	33.0	31.4	29.4	25.5	22.7
	2%	WB	17.4	18.8	20.5	22.6	24.7	25.6	25.2	25.1	24.9	23.9	20.8	18.1
		MCDB	20.5	22.3	24.9	28.3	30.7	32.2	32.4	32.6	30.6	27.9	23.8	20.6
	5%	WB	15.7	17.7	19.7	21.9	24.0	25.1	24.7	24.6	24.3	23.2	19.7	16.6
		MCDB	19.3	21.5	23.9	27.4	30.0	31.6	31.7	31.9	29.8	27.4	22.7	19.5
	10%	WB	13.8	16.4	18.8	21.2	23.4	24.5	24.2	24.2	23.9	22.5	18.6	14.7
		MCDB	17.9	20.7	23.4	26.4	29.4	30.9	31.0	31.2	29.3	26.7	22.2	18.2
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.8	12.5	12.7	11.6	11.4	11.4	11.5	10.9	11.2	11.4	12.3
		MCDBR	16.9	16.5	15.5	15.2	14.2	13.1	12.8	12.7	12.7	12.5	13.6	15.6
		MCWBR	8.8	7.9	6.7	6.4	5.0	3.9	3.9	3.6	4.1	5.0	6.8	8.1
	5% WB	MCDBR	12.4	11.5	11.1	11.3	10.8	10.9	11.2	11.3	10.0	9.2	9.9	11.2
		MCWBR	7.8	7.1	5.9	5.6	4.4	3.8	3.5	3.4	3.5	4.2	6.2	7.5
Clear-Sky Solar Irradiance	taub	0.299	0.304	0.325	0.372	0.393	0.402	0.412	0.393	0.385	0.344	0.318	0.301	
	taud	2.558	2.551	2.503	2.384	2.382	2.400	2.408	2.463	2.427	2.539	2.536	2.552	
	Ebn at Noon	948	970	964	922	898	884	874	890	890	913	916	927	
	Edh at Noon	85	93	104	122	122	120	118	111	111	93	86	82	
All-Sky Solar Radiation	RadAvg	3.32	4.09	5.12	5.96	6.18	6.91	6.83	6.52	5.36	4.39	3.50	3.02	
	RadStd	0.37	0.55	0.45	0.55	0.57	0.51	0.58	0.37	0.45	0.58	0.43	0.38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page